

Final Analytical Report

Flipkart E-Commerce Sales Dataset

January 2015 – December 2020

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Tool: Python | Pandas | Matplotlib | Seaborn | Jupyter Notebook



1. Title and Objective

This report presents the findings of a structured data science analysis performed on the Flipkart E-Commerce Sales Dataset as part of the WelIntern Pvt Ltd Week 1 Internship Assignment.

The primary objective of this assignment was threefold:

- Clean and validate the raw transactional dataset to ensure it is ready for accurate analysis.
- Compute essential business KPIs and identify meaningful sales and customer trends.
- Communicate findings through professional visualizations and a structured written report.

The analysis covers 113,000 sales transactions recorded between January 2015 and December 2020, spanning five product categories across multiple locations in Ghana.

2. Dataset Overview

Column Descriptions

Column	Type	Description
OrderDate	Date	Date the order was placed
OrderID	Integer	Unique identifier for each order
Delivery Date	Date	Date the order was delivered or returned
CustomerID	Integer	Unique identifier for each customer
Customer Age	Integer	Age of the customer (17–87)
Customer Gender	String	M (Male) or F (Female)
Location	String	City or region of the customer
Zone	String	Delivery zone classification (Zone 1–4)
Delivery Type	String	Express, Standard Delivery, or Shipped from Abroad
Product Category	String	Top-level product group
SubCategory	String	More specific product group
Product	String	Full product name
Unit Price	Integer	Base price of the product

Column	Type	Description
Shipping Fee	Integer	Cost charged for delivery
Order Quantity	Integer	Number of units ordered
Sale Price	Integer	Final price paid for the order
Status	String	Delivered or Returned
Reason	String	Return reason (blank if Delivered)
Rating	Integer	Customer satisfaction rating (1–5)

3. Data Cleaning Summary

The raw dataset was inspected systematically before analysis. The cleaning process followed four stages: null value review, duplicate detection, categorical standardization, and numerical validation.

3.1 Missing Value Analysis

A null value audit was performed on all 19 columns.

Column	Missing Count	Missing %	Action Taken
Reason	83,355	73.8%	Retained — blank means the order was Delivered (not returned). No fill needed.
All Other Columns	0	0%	No action required — all fields are fully populated.

3.2 Duplicate Records

A full-row duplicate check was performed using all 19 columns as keys.

Duplicate Check Result Total duplicate rows detected: 0 Total unique OrderIDs: 113,000 — matching total rows exactly. Conclusion: The dataset contains no duplicate entries. No records were removed.

3.3 Data Validation Checks

Validation Rule	Result	Action
Order Quantity ≥ 0	PASSED — No negative quantities found	No action needed
Sale Price > 0	PASSED — Minimum price is GHS 150	No action needed
Unit Price > 0	PASSED — All unit prices are positive	No action needed
OrderDate format valid	PASSED — All dates parsed as datetime64	No action needed
Delivery Date format valid	PASSED — All delivery dates parsed correctly	No action needed
Status values consistent	PASSED — Only 'Delivered' and 'Returned' present	No action needed
Rating range (1–5)	PASSED — All ratings fall within valid range	No action needed
Gender values consistent	PASSED — Only 'M' and 'F' present	No action needed
Category names consistent	PASSED — 5 clean, consistent category names	No action needed

3.4 Cleaned Dataset

After completing all validation steps, the cleaned dataset was saved separately:

- Raw file preserved at: data/raw/Flipkart_Sales_Dataset_raw_data_.xlsx
 - Cleaned file saved at: data/cleaned/flipkart_cleaned.csv
 - Total rows after cleaning: 113,000 (no records removed — dataset was already high quality)
 - Total columns: 19 (no columns dropped)
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4. KPI Summary

The following Key Performance Indicators were computed from the cleaned dataset. These metrics provide a high-level view of Flipkart's business performance over the 6-year period.

GHS 25,421,222 TOTAL REVENUE All completed & returned orders	113,000 TOTAL ORDERS Unique order transactions
GHS 225.0 AVG ORDER VALUE Revenue ÷ total orders	602,834 TOTAL UNITS SOLD Sum of all order quantities

KPI	Value	Interpretation
Total Revenue	GHS 25,421,222	Combined sale price of all 113,000 orders over 6 years
Total Orders	113,000	Unique order transactions recorded in the dataset
Average Order Value	GHS 225.0	Typical spend per transaction — consistent across the period
Total Units Sold	602,834	Total product units dispatched across all orders
Top Revenue Category	Health & Beauty	Contributed GHS 8,094,687 (31.8% of total revenue)
Second Category	Fashion	Contributed GHS 7,502,350 (29.5% of total revenue)
Order Delivery Rate	73.0%	82,474 out of 113,000 orders successfully delivered
Return Rate	27.0%	30,526 orders returned — a significant business concern
Average Customer Age	35.9 years	Prime working-age demographic drives most orders
Average Rating	2.7 / 5	Below average — suggests room for improvement in satisfaction

5. Trend Analysis

5.1 Monthly Sales Revenue Trend

Monthly revenue was aggregated over the 72-month period from January 2015 to December 2020. Revenue figures ranged from approximately GHS 306,000 to GHS 344,566, showing a relatively stable pattern throughout the period.

Key Trend Observations

December 2020 recorded the highest monthly revenue at GHS 344,566, aligning with the typical year-end holiday shopping surge.

February 2015 recorded the lowest monthly figure at GHS 306,329. This is consistent with seasonal slowdowns following January post-holiday spending reductions.

Revenue remained remarkably stable across the 6-year window, with no dramatic crashes or spikes. Month-to-month variation stayed within a ~GHS 40,000 band, suggesting consistent operational performance.

5.2 Revenue by Product Category

Revenue contribution was analyzed across all five product categories for the full period.

Rank	Product Category	Total Revenue (GHS)	Share of Total Revenue
1	Health and Beauty	8,094,687	31.8%
2	Fashion	7,502,350	29.5%
3	Phones and Tablet	4,043,431	15.9%
4	Home and Office	3,466,919	13.6%
5	Electronics	2,313,835	9.1%

Health and Beauty and Fashion together represent over 61% of all revenue. This strong concentration means Flipkart's overall performance is heavily dependent on these two categories. Electronics, while the lowest contributor, may have higher per-unit margins and growth potential.

5.3 Revenue by Delivery Zone

Sales were mapped to four delivery zones based on customer location.

Zone	Total Revenue (GHS)	Share	Key Locations
Zone 3	10,966,263	43.1%	Ashanti, Western, Volta — largest revenue zone
Zone 1	6,403,235	25.2%	Greater Accra — capital region
Zone 2	4,667,420	18.4%	Weija, Oyarifa, Amasaman
Zone 4	3,384,304	13.3%	Upper East, Upper West, Oti, Northern

Zone 3 dominates total revenue despite not containing the capital. This suggests that Ashanti and Western regions have strong purchasing power and high order volumes. Zone 4, covering northern and eastern periphery regions, contributes the least — possibly due to limited access or lower purchasing capacity.

5.4 Delivery Type Performance

Revenue was distributed almost equally across all three delivery types, indicating no single method dominates customer preference.

Delivery Type	Total Revenue (GHS)	Order Count (approx.)
Shipped from Abroad	8,514,130	37,853
Standard Delivery	8,512,908	37,841
Express	8,394,184	37,306

The near-equal split across delivery types suggests customers are price-sensitive: Express delivery costs more but is chosen by roughly a third of customers for speed. Promoting Standard Delivery with a small discount could shift high-volume customers toward lower operational cost per order.

6. Customer Insights

6.1 Customer Age Distribution

Customer ages ranged from 17 to 87, with a mean age of 35.9 years and a median of 35. The age distribution reveals which segments drive order volume.

Age Group	Orders	Share of Orders	Insight
18–25	20,868	18.5%	Younger shoppers — likely price-conscious and fashion-driven
26–35	39,566	35.0%	Largest segment — active professionals with stable income
36–50	39,543	35.0%	Second largest — family-oriented purchases dominate
50+	13,023	11.5%	Smaller but potentially high-value segment

The 26–35 and 36–50 age groups together account for 70% of all orders. These are prime working-age adults who likely shop for household, fashion, and health products regularly. Marketing campaigns targeting this demographic with subscription deals or loyalty rewards could significantly increase retention.

6.2 Customer Gender Split

The gender distribution is nearly balanced, with a slight lean toward male customers.

Gender	Order Count	Share
Male (M)	58,287	51.6%
Female (F)	54,713	48.4%

The near-equal gender split suggests the platform appeals broadly rather than skewing toward one segment. However, given that Fashion and Health & Beauty are the top revenue categories — products typically associated with female purchasing patterns — there may be an untapped opportunity to increase female-targeted marketing to grow the female customer base further.

6.3 Customer Rating Analysis

Customer ratings (1–5 scale) were analyzed to gauge satisfaction levels. The distribution reveals a concerning pattern.

Rating	Count	Interpretation
1 (Very Poor)	26,661	23.6% — highest frequency of any rating
2 (Poor)	26,703	23.6%
3 (Neutral)	26,624	23.6%
4 (Good)	16,517	14.6%
5 (Excellent)	16,495	14.6%

6.4 Return Reason Analysis

Among the 30,526 returned orders, the following return reasons were most commonly cited:

Return Reason	Count	% of Returns	Business Implication
Quality — Defective item	6,065	19.9%	Product quality control failure
Delivery — Missing item/part	6,013	19.7%	Packaging or fulfillment error
Product — Not fitting expectation	5,872	19.2%	Product description mismatch
Delivery — Wrong item	5,868	19.2%	Warehouse picking or dispatch error
Onsite — Description mismatch	5,827	19.1%	Website listing inaccuracy

Remarkably, the five return reasons are distributed almost equally, each accounting for approximately 19–20% of returns. This even spread suggests that returns are not caused by one isolated failure — they reflect systemic issues across product quality, fulfillment accuracy, and listing integrity. A targeted improvement in any one of these areas could reduce the overall return rate by approximately 5%.

7. Visualization Section

Four visualizations were created for this analysis. Each chart is saved in the outputs/ directory of the project repository. A brief description and interpretation is provided for each.

Chart 1 — Bar Chart: Revenue by Product Category

Chart Details

File: outputs/charts/revenue_by_category.png

Type: Horizontal / Vertical Bar Chart | Tool: Matplotlib + Seaborn

X-axis: Product Category | Y-axis: Total Revenue in GHS

Interpretation: Health and Beauty leads with GHS 8.09M, followed closely by Fashion at GHS 7.50M. These two categories together form the commercial backbone of the platform, contributing 61.3% of total revenue. Electronics, despite being a premium category, contributes only 9.1%, suggesting either lower order volume or lower average pricing relative to Health and Beauty products. Flipkart should consider bundling or cross-promotion strategies between Health and Electronics to lift the lower-performing category.

Chart 2 — Line Chart: Monthly Sales Revenue Trend

Chart Details

File: outputs/charts/monthly_sales_trend.png

Type: Line Chart with markers and area fill | Tool: Matplotlib

X-axis: Month (Jan 2015 – Dec 2020) | Y-axis: Monthly Revenue in GHS

Interpretation: The monthly revenue line shows a stable, consistent pattern over the entire 6-year period. There is no dramatic growth trajectory, which may indicate that the platform has reached a steady-state in its current markets. December months consistently show slight revenue elevations, consistent with end-of-year gifting and holiday shopping behavior. The absence of major peaks or troughs also suggests resilient demand that is not heavily affected by external shocks — a positive sign for operational planning.

Chart 3 — Pie Chart: Delivery Status Distribution

Chart Details

File: outputs/charts/delivery_status_pie.png

Type: Pie Chart with exploded slices | Tool: Matplotlib

Segments: Delivered (73.0%) vs Returned (27.0%)

Interpretation: The pie chart reveals that 27% of all orders — more than 1 in 4 — are returned. This is an elevated return rate by e-commerce standards. A globally healthy return rate typically sits between 8–15%. Flipkart's 27% return rate signals deep issues in either product quality, listing accuracy, or fulfillment operations. Reducing returns by even 10 percentage points (to ~17%) would recover millions in lost revenue and significantly improve customer satisfaction scores.

Chart 4 — Dashboard Mockup

Chart Details

File: outputs/dashboard/dashboard_mockup.png

Type: Multi-panel static dashboard | Tool: Matplotlib GridSpec

Panels: KPI Cards, Monthly Trend, Category Revenue, Delivery Status, Top 5 Products

Interpretation: The dashboard presents all major KPIs and charts on a single canvas. KPI cards at the top provide immediate visibility of Total Revenue, Total Orders, Average Order Value, and Units Sold. The trend chart confirms revenue stability. The horizontal bar chart quickly identifies Health and Beauty and Fashion as revenue leaders. Together, the dashboard enables a non-technical stakeholder to grasp the full state of Flipkart's sales performance in under one minute.

8. Key Findings

The following findings are drawn directly from the data and represent the most important discoveries of this analysis:

Finding 1 — Revenue is Dominated by Two Categories

Health and Beauty (GHS 8.09M) and Fashion (GHS 7.50M) account for 61.3% of total revenue. The remaining three categories — Phones & Tablet, Home & Office, and Electronics — share only 38.7% of revenue. Category diversification efforts should focus on growing Electronics and Home & Office, which have clear upsell potential.

Finding 2 — Return Rate of 27% Represents a Major Revenue Leak

Over 30,526 orders were returned, representing GHS 6.86M+ in potential revenue that was not retained. The five return reasons are distributed almost equally, pointing to multi-system failure rather than a single isolated problem. Addressing product quality, fulfillment accuracy, and listing correctness simultaneously is critical.

Finding 3 — Revenue is Stable but Not Growing

Monthly revenue showed minimal growth over six years (2015–2020), with values consistently between GHS 306,000 and GHS 345,000. While stability is operationally positive, the lack of growth may suggest the platform has plateaued. Expansion into untapped zones (Zone 4 regions) or new product categories could reignite growth.

Finding 4 — Customer Satisfaction is Below Average

The average rating of 2.7 out of 5 and the fact that over 70% of customers gave ratings of 3 or below indicate widespread dissatisfaction. Since ratings correlate strongly with returns, improving the return drivers will likely also raise ratings and repeat purchase rates.

Finding 5 — Zone 3 Drives the Most Revenue Despite Not Being the Capital Zone

Zone 3 (Ashanti, Western, Volta) contributes 43.1% of total revenue — significantly more than Zone 1 (Greater Accra, 25.2%). This counterintuitive finding suggests that Ashanti region in particular has a highly active and loyal customer base. Tailored regional promotions for Zone 3 customers could deepen this advantage.

Finding 6 — The 26–50 Age Group Drives 70% of Orders

Customers aged 26–50 placed 79,109 orders out of 113,000 — a 70% majority. This working-age segment is the primary revenue driver and should be the primary target for loyalty programmes, subscription offers, and personalised product recommendations.

9. Recommendations

Based on the findings above, the following data-driven recommendations are offered to improve Flipkart's business performance:

Priority	Recommendation	Expected Impact
HIGH	Implement a Product Quality Audit for the top-returned subcategories (Medical Equipment, Fashion footwear). Establish quality checkpoints before dispatch.	Reduce return rate from 27% to below 20%, recovering an estimated GHS 3M+ in annual revenue
HIGH	Fix product listing descriptions for Electronics and Home & Office categories. Hire a content QA team to verify photos and specs match physical products.	Reduce 'Description mismatch' and 'Not fitting expectation' return reasons — each account for ~19% of returns
HIGH	Improve fulfillment accuracy by auditing warehouse picking processes. Missing item and wrong item returns each account for ~19% of returns.	Direct reduction of fulfillment-related returns — approximately 12,000 orders annually
MEDIUM	Launch a customer loyalty programme targeting the 26–50 age group with points, early access, and free express shipping above a threshold order value.	Increase repeat purchase rate and average order value from GHS 225

Priority	Recommendation	Expected Impact
MEDIUM	Develop targeted marketing campaigns for Zone 4 (Northern Ghana) where revenue is currently lowest at 13.3%. Consider lower shipping fees or local partnerships.	Expand the active customer base into an underserved high-population region
LOW	Investigate growth opportunities in Electronics — currently the lowest revenue category at 9.1% but a premium segment. Bundle electronics with accessories.	Potential to double Electronics revenue with focused cross-selling
